

a

Dataset evidence-gap profile

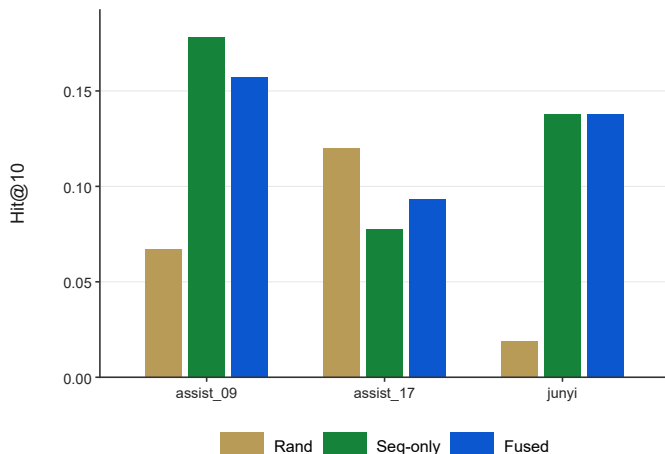
Percentages are shown explicitly; history is median interactions.

assist_09	82.8% Single	0.9% Item edge	64.3% Seq edge	3.1% Unseen	3.1% Bridge	41 Hist med
assist_17	78.3% Single	6.2% Item edge	76.3% Seq edge	2.8% Unseen	2.8% Bridge	148 Hist med
junyi	100.0% Single	0.0% Item edge	25.2% Seq edge	100.0% Unseen	100.0% Bridge	25 Hist med

b

History-to-query retrieval

Direct-unseen bridgeable query concepts.



c

Held-out transition retrieval

Global route recovery controls.

